

Metadata

Aloha name

Author

Date

Universe

Neutralization

Decay

Truncation

Alpha Design Steps

Step	Guiding prompts	Your answer
1		
2		
3		
4		
5		

Robustness & Final Alpha

Robustness: group-rank, group-rank, neutralization.
Write the base formula before transformations.
Specify lookback windows, decay, truncation.
Key risks or failure cases.
Expected direction: high = long or short?
Suggested robustness checks (subperiods, sectors).
Final alpha formula (pseudo-code) and rationale.

1. Objective & Edge

Short-term price dislocations arise from liquidity shocks and investor overreaction. These effects are temporary and tend to mean-revert.

2. Hypothesis

Stocks with extreme negative short-term returns will outperform, while extreme positive return stocks will underperform.
Expected holding horizon: 2–10 trading days.

3. Mathematical Expression

Raw variable: daily returns.
Compute cumulative returns over a short window (e.g. 5 days).

4. Transform & Refine

Apply cross-sectional z-score to identify extremes.
Rank within industry to control sector exposure.
Low z-score = Long, High z-score = Short.

5. Final Alpha Formula

Pseudo-code:
`group_rank( -zscore(ts_sum(returns, 5)), industry )`

Rationale: exploits short-term mean reversion while neutralizing industry effects.

Step					Assumpt